

Self-Reflective Report

BIT1123 Object Oriented Programming

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Class Code	202605F0783
Programme	Software Engineering
NRIC / Passport	To be added by the student before LMS submission
Repository URL	https://github.com/X2bl/JIHA_OMAR_202501010623_202605F0783_-BIT1123_Object_Oriented_Programming

1. Introduction

This report explains my learning journey in BIT1123 Object Oriented Programming. At the beginning, Java was new for me. By doing the tutorials one by one, I started to understand how Java programs are written and how OOP ideas are used.

2. Knowledge Gained from Each Tutorial

Week 1: I learned how to use GitHub, create a Java file, and run a simple Hello World program. I also made a simple program to display a student grade.

Week 2: I learned how to create a class, add attributes, make a constructor, create an object, and call methods.

Week 3-4: I learned about inheritance. The Student and Lecturer classes use the Person class. I also learned method overriding and polymorphism when the introduce() method gives different output.

Week 5: I learned encapsulation by using private variables with getters and setters.

Week 6: I learned more about inheritance by making Lecturer a child class of Employee.

Week 7: I learned abstraction by making an abstract Appliance class. I created WashingMachine and Refrigerator classes that inherit from it. Each appliance has its own operate() method, but they use the same common methods for the brand and power.

Week 8-9: I learned how to use ArrayList, Scanner, and text files to save and read tasks.

Week 10: I learned basic Java Swing GUI programming, buttons, and action events.

3. Challenges Encountered

At first, I found it difficult to understand the difference between a class and an object. I also made small mistakes with constructors, method names, and Java file names. Another challenge was arranging the folders in GitHub in a clear way.

4. How I Overcame the Challenges

I solved these problems by reading the tutorial examples again and writing small programs. I checked my spelling carefully and used the same name for a public class and its Java file. I also used clear week folders and a README file so the repository is easier to read.

5. Improvement in My Java Programming Skills

My Java skills improved because I can now write a basic class, create objects, use variables, constructors, methods, and simple input. I can also read simple error messages and make corrections. I still need more practice, but I am more confident than when I started.

6. My Understanding of OOP Concepts

I understand that a class is like a plan, while an object is made from that plan. Encapsulation keeps data safe by using private variables. Inheritance lets one class reuse another class. Polymorphism means the same method can work in different ways, such as the introduce() method in the Person, Student, and Lecturer classes.

7. Future Learning Plans

I want to practise more Java exercises, especially abstraction and inheritance. I also want to learn more about arrays, exception handling, files, and GUI applications. In the future, I will use GitHub more often to keep my projects organised.

8. Conclusion

Overall, this subject helped me learn Java and the basic concepts of OOP. The tutorials were useful because they gave me small tasks to practise. I will continue practising so I can write better and more organised programs.

9. GitHub Repository URL

https://github.com/X2bl/JIHA_OMAR_202501010623_202605F0783_-BIT1123_Object_Oriented_Programming